Review: Exponents and radicals

1 Explain how to evaluate 4^5.

2 Explain how to determine if the answer is positive or negative when evaluating an expression with an integer power and negative base such as $(-7)^3$.

3 By considering the expressions $(-3)^2$ and -3^2, state whether the each of the following statements is correct.

a Without parentheses, the base is -3 and the power is 2. The result is 9.

b If you have parentheses surrounding the -3, then the base is 3 and the power is 2. You first evaluate to obtain 9, then take the negative, which is -9.

c Without parentheses, the base is 3 and the power is 2. You first evaluate 3^2 to obtain 9, then take the negative, which is -9.

d If you have parentheses surrounding the -3, then the base is -3 and the power is 2. The result is 9.

4 $(-5)^{13}$ simplifies to which of the following?

A 5^{-13} **B** -5^{13} **C** -5^{-13} **D** 5^{13}

5 Evaluate:

a 3^1

b 1^3

c 2^6

d $(-2)^4$

e $3^3 \times 3^2$

f $(-4)^3$

g $(-2)^3 \times 3^4$

h -6^3

i $7^2 - 2^2$

j $(-11)^2 - (-9)^2$

k $(-3)^2 \times (-2)^2$

l -4^3

m $(-3)^4$

n -3^4

o $(-4)^2$

p $(-16)^2$

q -15^2

r $-(-8)^2$

s $(-9) \times (-6)^2$

t $(-2)^3 + 5^2$

u $4 - (5 - 2)^2$

v $6 - \left(12 - 3^2\right)$

w $9 - 2^2 \times (-7)$

x $(2 - 5)^2 \times (-21 + 14)$

y $-(-3)^3 + 8^2$

z $-(-4)^3 - (-10)^2 - 122$

6 A square patch of grass has a side length of  Write an expression for the area of the grass in exponential form.

7 Determine whether each of the following statements is true or false.

a $-x^n = (-x)^n$ for all x, only if n is odd.

b $-x^n = (-x)^n$ for all x always.

c $-x^n = (-x)^n$ for all x, only if n is even.

8 Evaluate the following:

a $\sqrt{4}$
b $\sqrt{16}$
c $\sqrt{64}$
d $\sqrt{100}$
e $\sqrt{144}$
f $\sqrt{25} - \sqrt{9}$
g $\sqrt{8^2 + 6^2}$
h $\sqrt{10^2 - 6^2}$
i $\sqrt{144} - \sqrt{16}$
j $\sqrt{16}$
k $\sqrt{100}$
l $\sqrt{169}$
m $\sqrt{225}$
n $-\sqrt{4}$
o $-\sqrt{81}$
p $-\sqrt{121}$
q $-\sqrt{196}$

Additional questions

9 For each of the following, determine the square root.

a We know that $5 \times 5 = 25$. Find $\sqrt{25}$.

b We know that $7 \times 7 = 49$. Find $\sqrt{49}$.

c We know that $6 \times 6 = 36$. Find $-\sqrt{36}$.

d We know that $7 \times 7 = 49$. Find $-\sqrt{49}$.

10 For each of the following, determine the cube root.

a We know that $2 \times 2 \times 2 = 8$. What does $\sqrt[3]{8}$ equal?

b We know that $4 \times 4 \times 4 = 64$. What does $\sqrt[3]{64}$ equal?

c We know that $-3 \times -3 \times -3 = -27$. What does $\sqrt[3]{-27}$ equal?

d We know that $-5 \times -5 \times -5 = -125$. What does $\sqrt[3]{-125}$ equal?

11 Evaluate the following:

a $\sqrt[3]{8}$
b $\sqrt[3]{64}$
c $\sqrt[3]{125}$
d $\sqrt[3]{27}$

12 Which is bigger: 3^3 or 5^2?

13 Evaluate the following expressions:

a $\left(2^2+3^3\right)$

b $(3+4)^2$

c $2^3\,(3+7)$

d $\left(\sqrt{9}+1\right)^3$

e $\left(\sqrt{3^2+4^2}\right)^3$

f $\dfrac{2^3+\sqrt{16}}{4}$

g $\sqrt{64}\div\sqrt[3]{8}$

h $2\left(\sqrt[3]{8}+4\right)^2$

i $\left(\dfrac{1}{2}\right)^2+\dfrac{1}{2}$

j $2\left(\dfrac{2}{3}\right)^2$

k $\left(\dfrac{2}{4}+\dfrac{1}{2}\right)^2$

l $\left(\dfrac{2}{3}+\sqrt{\dfrac{1}{9}}\right)^3$

m $\sqrt{1-\left(\dfrac{6}{10}\right)^2}$

n $\left(-\dfrac{2}{3}\right)^2+\sqrt{\dfrac{25}{81}}$

o $\sqrt{81}\div\sqrt[3]{125}$

p $2\left(\dfrac{1}{\sqrt[3]{8}}+4\right)$

14 For each of the following situations:

i Determine the error each student made.

ii Write the correct work and answer.

a Ralph tried to evaluate the expression $2\,(3+2)^2$. He thought that he had the right answer and submitted the following to his teacher $2\,(3+2)^2=2\,(5)^2=10^2=100$.

b Ida tried to evaluate the expression $\sqrt{9}\left(8\div 4-3\times 4+4^2+3\right)$. She submitted the following to her teacher $\sqrt{9}\left(8\div 4-3\times 4+4^2+3\right)=\sqrt{9}\,(2-12+16+3)=\sqrt{9}\,(9)=\sqrt{81}=9$.